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Code

JupyterLab Python 3 (ipykernel)

[2]:

```
# ============================================================
# EXPERIMENT 3(B)
# BIVARIATE ANALYSIS:
# LINEAR AND LOGISTIC REGRESSION MODELING
# ============================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import (
    r2_score,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    mean_squared_error,
    mean_absolute_error,
    confusion_matrix
)

print("=" * 75)
print("EXPERIMENT 3(B) - BIVARIATE ANALYSIS")
print("=" * 75)


# ============================================================
# 1. CREATE UCI DIABETES SAMPLE DATASET
# ============================================================

uci_data = {
    "Glucose": [
        90, 162, 184, 119, 175,
        137, 145, 111, 120, 130,
        150, 160, 125, 140, 180,
        100, 155, 135, 170, 110
    ],

    "BloodPressure": [
        72, 61, 76, 83, 101,
        82, 85, 63, 80, 78,
        75, 70, 80, 85, 90,
        65, 88, 82, 92, 70
    ],

    "SkinThickness": [
        16, 34, 25, 20, 30,
        29, 30, 39, 25, 35,
```

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Code JupyterLab Python 3 (ipykernel)

```
# ============================================================
# 2. CREATE PIMA INDIANS DIABETES SAMPLE DATASET
# ============================================================

pima_data = {
    "Glucose": [
        81, 172, 90, 187, 133,
        136, 135, 122, 140, 155,
        100, 165, 120, 145, 190,
        85, 150, 130, 175, 110
    ],

    "BloodPressure": [
        97, 50, 84, 94, 94,
        82, 82, 52, 80, 90,
        70, 65, 75, 85, 95,
        68, 88, 80, 92, 72
    ],

    "SkinThickness": [
        15, 24, 40, 30, 46,
        29, 30, 17, 25, 35,
        20, 32, 28, 35, 40,
        18, 31, 27, 34, 22
    ],

    "Insulin": [
        289, 229, 136, 288, 23,
        148, 152, 168, 125, 200,
        100, 220, 130, 180, 270,
        70, 160, 140, 230, 95
    ],

    "BMI": [
        27.06, 18.59, 32.97, 38.22, 37.24,
        32.48, 32.63, 18.59, 30.50, 34.20,
        25.40, 36.20, 28.50, 31.80, 40.10,
        23.70, 34.60, 30.20, 37.80, 27.90
    ],

    "DiabetesPedigreeFunction": [
        0.79, 0.65, 0.95, 1.03, 1.49,
        1.23, 1.21, 0.12, 0.50, 0.70,
        0.40, 1.10, 0.60, 0.80, 1.40,
        0.20, 0.90, 0.55, 1.25, 0.35
    ],

    "Age": [
        62, 21, 29, 35, 62,
        50, 50, 26, 45, 55,
        28, 47, 33, 52, 65,
        24, 43, 38, 60, 31
```

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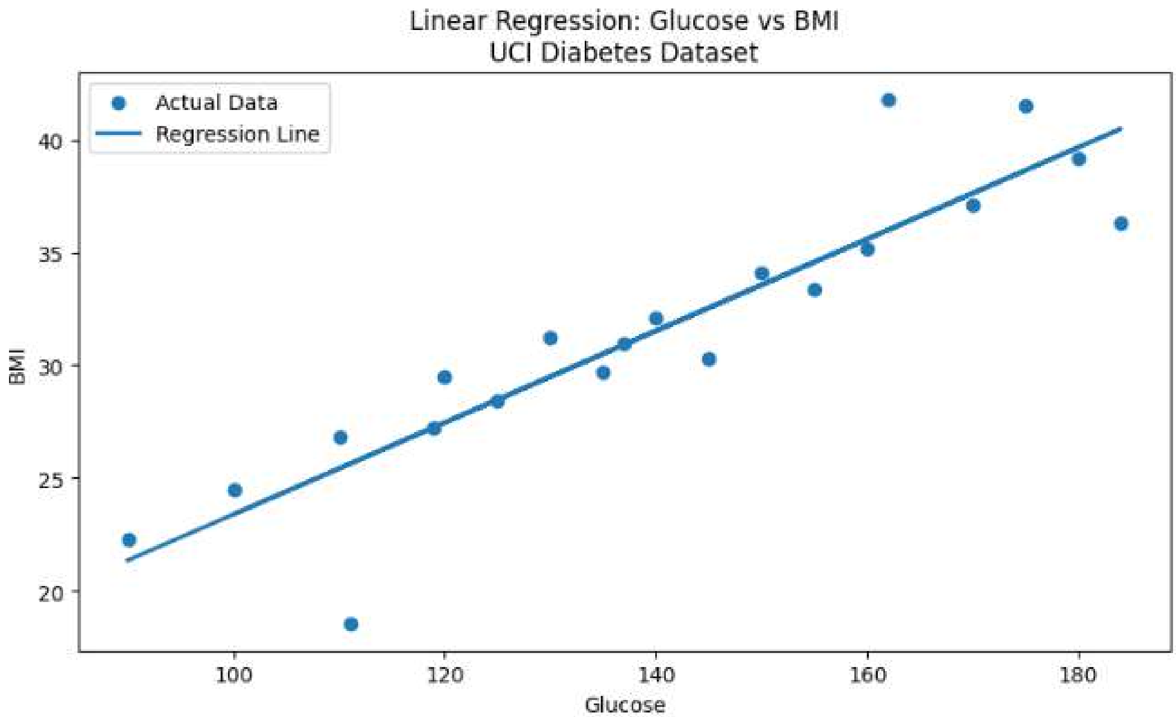
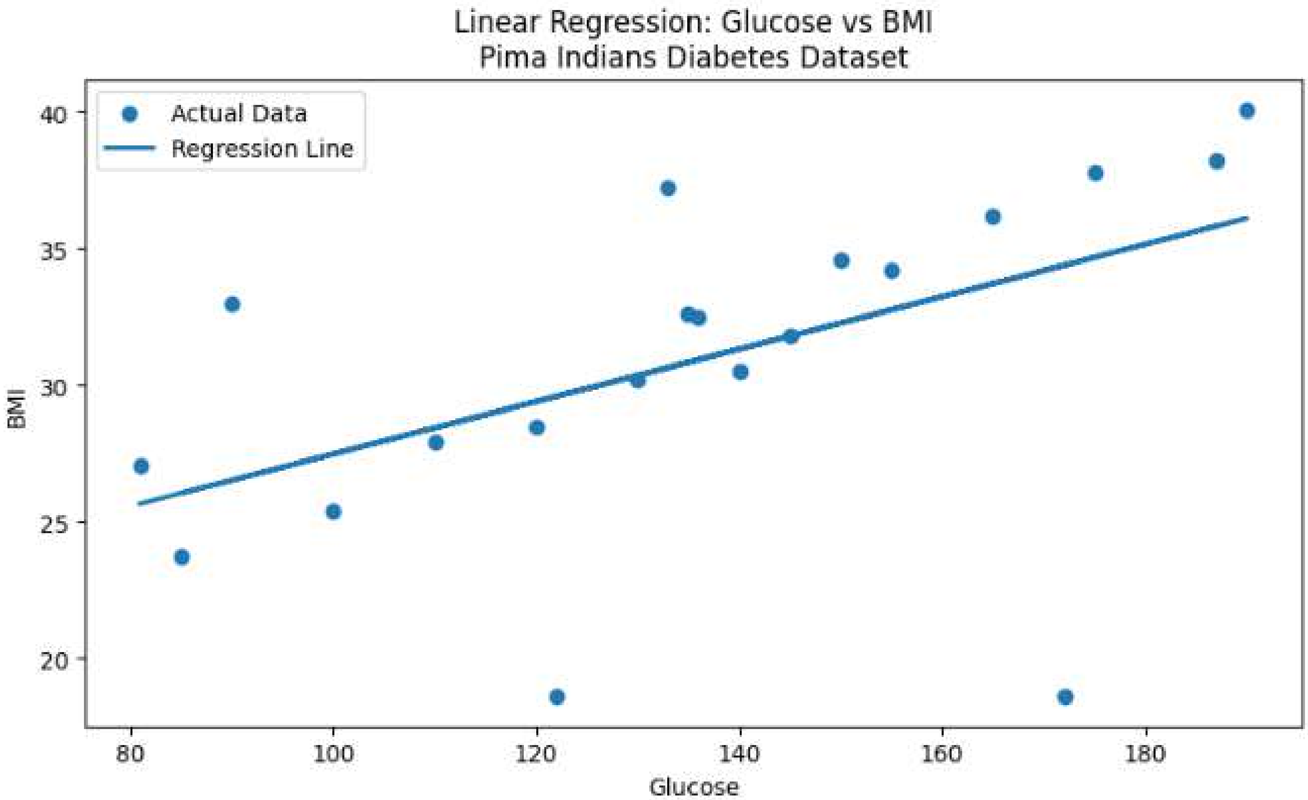
Code JupyterLab Python 3 (ipykernel)

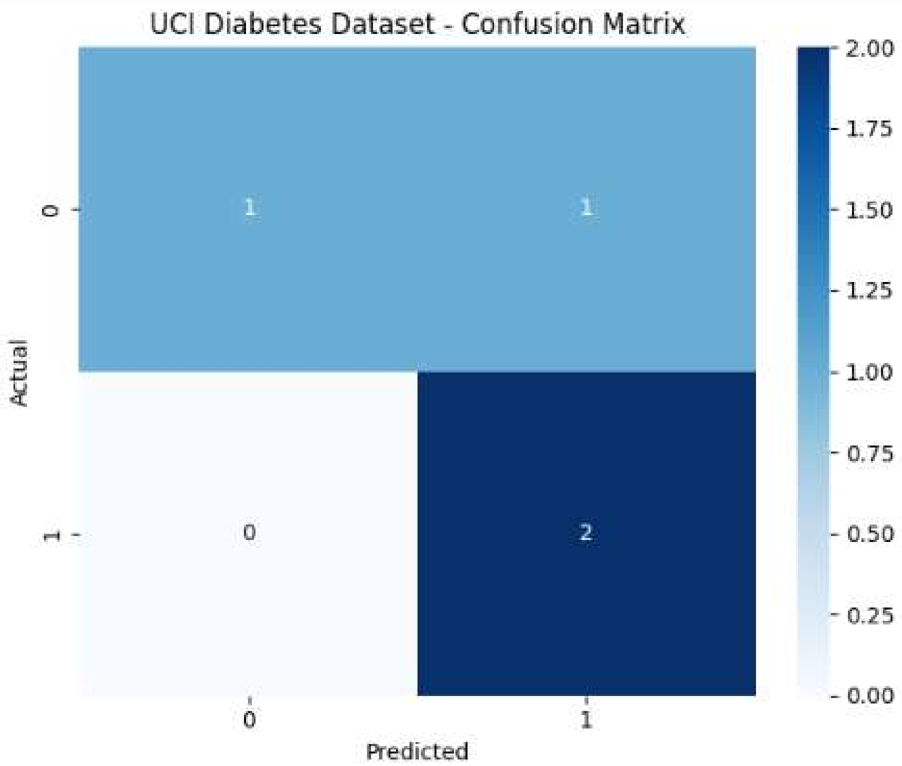
```python
# ============================================================
# 7. FINAL RESULT
# ============================================================

print("\n" + "=" * 75)
print("RESULT")
print("=" * 75)

print(
    "Linear Regression reveals the relationship between "
    "Glucose Level and BMI."
)

print(
    "Logistic Regression predicts diabetes presence using "
    "Glucose, BloodPressure, BMI, and Age."
)

print(
    "R² Score is used to evaluate Linear Regression, while "
    "Accuracy, Precision, Recall, and F1 Score are used to "
    "evaluate Logistic Regression."
)

print(
    "The results demonstrate differences in predictive "
    "performance between the UCI and Pima datasets."
)

print("=" * 75)
print("EXPERIMENT 3(B) COMPLETED SUCCESSFULLY")
print("=" * 75)
```

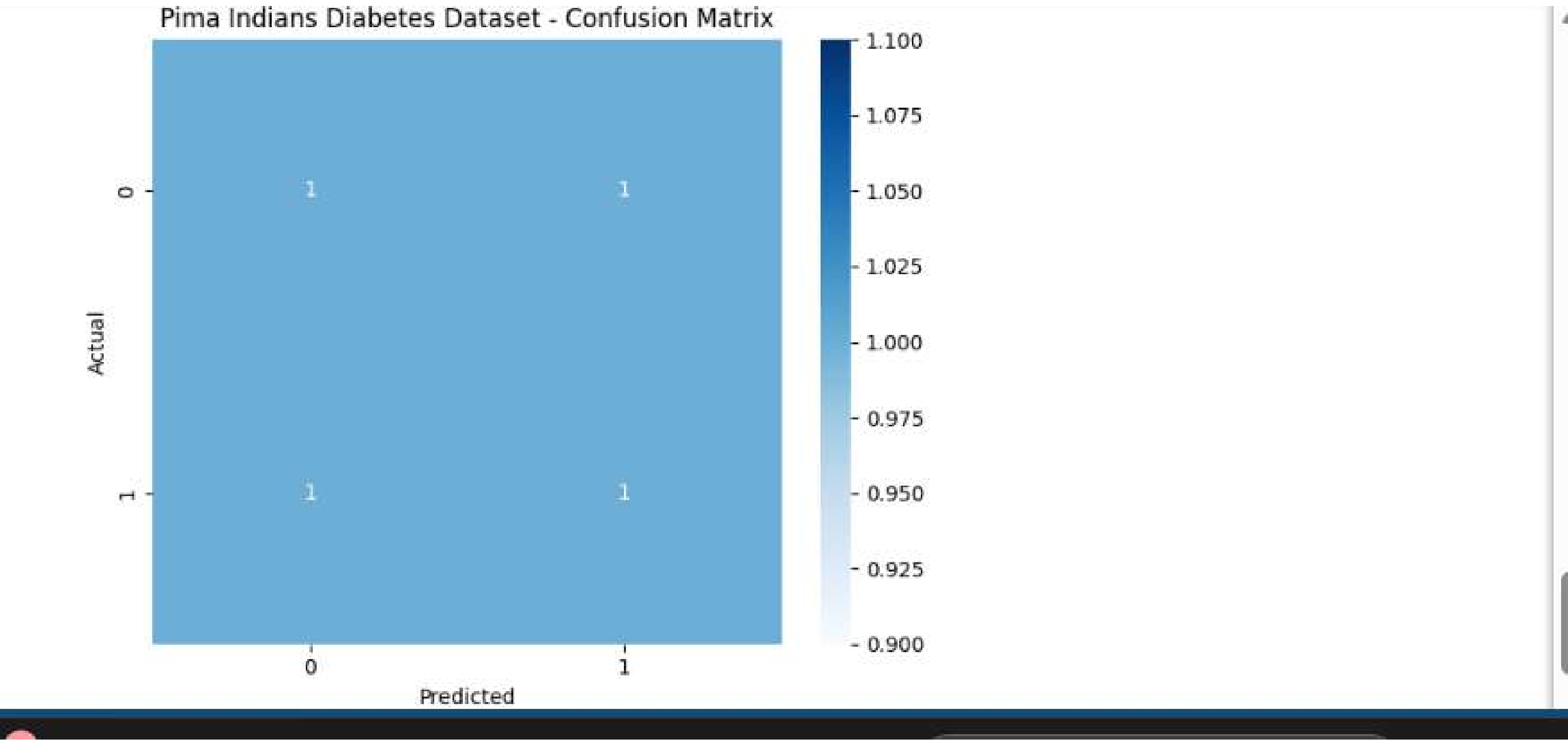
```
===========================================================================
EXPERIMENT 3(B) - BIVARIATE ANALYSIS
===========================================================================

===========================================================================
UCI DIABETES DATASET SAMPLE
===========================================================================
   Glucose  BloodPressure  SkinThickness  Insulin    BMI  \
0       90             72             16      264  22.27
1      162             61             34      143  41.81
2      184             76             25      128  36.31
3      119             83             20      230  27.26
4      175            101             30       63  41.49

   DiabetesPedigreeFunction  Age  Outcome
0                      1.16   79        0
1                      2.15   35        0
2                      1.41   31        0
```



Linear Regression: Glucose vs BMI
UCI Diabetes Dataset
Actual Data
Regression Line
BMI
Glucose
20
25
30
35
40
100
120
140
160
180


Linear Regression: Glucose vs BMI
Pima Indians Diabetes Dataset
Actual Data
Regression Line
40
35
30
25
20
BMI
80
100
120
140
160
180
Glucose


UCI Diabetes Dataset - Confusion Matrix
Actual
Predicted
0
1
1
1
0
2
0
1
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00



Pima Indians Diabetes Dataset - Confusion Matrix
Actual
Predicted
0
1
1
1
1
1
1.100
1.075
1.050
1.025
1.000
0.975
0.950
0.925
0.900